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TITLE: Hierarchical Deep Reinforcement Learning for Safe Variable-Rate UAV Spraying in an Orchard Autonomy Simulator

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ABSTRACT

Autonomous agricultural spraying requires a planner that can cover target crops, satisfy variable-rate application demands, avoid static and moving obstacles, and generate missions that remain executable in a robotic simulator. This paper presents a hierarchical deep reinforcement learning framework for UAV spraying path planning in an orchard scenario derived from the Aerial Autonomy Stack (AAS). Rather than manually creating an abstract grid map, the framework parses the existing Gazebo orchard world, extracts trees, field regions, and obstacles, and converts them into a grid-based Markov decision process. The core contribution is not a new DQN architecture; instead, it is a task decomposition that connects RL motion selection, demand-aware automatic spraying, local safety control, and AAS mission export. We compare DQN-family methods, including DQN, Double DQN, Dueling DQN, Rainbow DQN lite, and DRQN, under static coverage, variable-rate spraying, dynamic-obstacle, and whole-farm coverage settings. The static tree-row experiments show that Dueling DQN is a stable baseline, while early enhanced experiments reveal that explicit spray switching greatly increases exploration difficulty. In the five-seed hierarchical enhanced task, DQN achieves the highest success rate of 80.0%, while DRQN obtains the highest mean demand satisfaction of 80.9% +/- 14.7%; all compared DQN-family methods report zero planning and dynamic collisions. Ablation experiments show that auto-spray is the main contributor to success and demand satisfaction, while the safety controller removes collision failures. For larger sparse whole-farm coverage, unguided reinforcement learning reaches only 15.9% coverage, whereas curriculum-guided DRQN reaches 100.0% success and 97.56% +/- 0.00% coverage over three seeds. Finally, a successful RL path is exported as an AAS mission YAML, establishing the interface needed for Gazebo replay validation. These results suggest that safe orchard spraying is better addressed as a hierarchical planning-and-control problem than as a purely end-to-end DQN task.

KEYWORDS

deep reinforcement learning, DQN, DRQN, UAV spraying, precision agriculture, coverage path planning, Gazebo, safety controller

SECTION: Introduction

Precision agriculture increasingly requires robots that can execute spatially selective operations rather than uniform field traversal. UAV spraying is a representative task: the vehicle must visit crop targets, satisfy application demand, avoid obstacles, reduce redundant spraying, and produce a mission that is compatible with a robotics autonomy stack. Rule-based coverage paths and nearest-target heuristics are simple and often effective in structured fields, but they are difficult to adapt when variable demand, dynamic obstacles, partial observability, and long-distance transfers must be optimized jointly.

Deep reinforcement learning (DRL) provides a natural modeling tool for such sequential decision problems. However, directly applying a single end-to-end DQN policy to every part of orchard spraying is not necessarily appropriate. The agent may need to learn navigation, spray activation, obstacle avoidance, and long-range exploration simultaneously, which enlarges the action space and makes rewards sparse. Our experiments indicate that this issue becomes severe when spraying is extended from a local tree row to whole-farm multi-block coverage.

This paper studies UAV spraying path planning using DQN-family algorithms in an AAS/Gazebo orchard environment. The central question is not whether a single DQN variant dominates all scenarios. Instead, we ask which algorithmic components are useful at different task levels, and whether a hierarchical design can make the enhanced spraying task safer and easier to learn. The resulting framework parses an existing orchard simulator, converts it into a grid decision environment, trains multiple value-based policies, evaluates them under multi-seed protocols, and exports successful paths into AAS mission YAML files.

The main contributions are:

We build an AAS-derived orchard spraying benchmark that converts Gazebo world information into a grid MDP with target trees, field regions, static obstacles, variable spraying demand, and dynamic safety values.

We introduce and evaluate a hierarchical spraying design that combines RL motion planning, automatic demand-aware spraying, and a local safety controller.

We compare DQN, Dueling DQN, Rainbow DQN lite, DRQN, PPO, and rule-based baselines across static coverage, variable-rate spraying, dynamic-obstacle, and whole-farm coverage settings.

We report five-seed enhanced-task results, ablations on auto-spray, safety control, recurrent memory, and 4-action versus 8-action spaces, together with reward, coverage, and demand-satisfaction learning curves where the logging protocol is complete.

We export a successful RL path into an AAS mission YAML and define the Gazebo/QGroundControl replay evidence needed to close the loop from offline planning to simulator execution.

SECTION: Related Work

SUBSECTION: Coverage Path Planning for Agricultural Robots

Coverage path planning aims to visit a target region or a set of targets while minimizing path length, overlap, or operation cost. Classical methods include boustrophedon decomposition, graph search, and heuristic nearest-target traversal [ref]. Recent UAV path-planning surveys emphasize that path length, flight time, energy, environmental uncertainty, and collision avoidance must be considered jointly [ref]. In agricultural contexts, crop-row extraction from UAV imagery has been used to support irrigation management and spatially selective field operations [ref], while UAV-based pesticide irrigation has been formulated as an optimization problem using adaptive ant colony optimization to reduce flight path, energy use, and pesticide residue [ref]. These methods are computationally efficient and interpretable, but they often require explicit cost design and may be less flexible when operational objectives such as variable-rate spraying and dynamic obstacle safety must be optimized together.

Informative path-planning methods have also been used for UAV target search in cluttered environments, where information gain, coverage, sensor performance, and collision avoidance must be traded off [ref]. Reinforcement learning has been explored for agricultural search and localization as well. Van Essen et al. learn UAV search policies for non-uniform weed localization using deep Q-learning and prior knowledge, showing that learned policies can reduce path length compared with row-by-row flight in non-uniform fields [ref]. Our work is similar in using a value-based RL planner for agricultural UAV routing, but differs by focusing on spray-demand satisfaction, safety overrides, dynamic obstacles, and export to an AAS/Gazebo mission format.

SUBSECTION: Deep Reinforcement Learning for Navigation

DQN introduced deep value learning for high-dimensional decision problems [ref]. Double DQN reduces overestimation bias [ref], Dueling DQN separates state value and action advantage [ref], and Rainbow combines several DQN improvements [ref]. DRQN adds recurrent memory to Q-learning, making it suitable for partially observable settings [ref]. These methods motivate our algorithm set. We focus on how DQN-family methods behave when the same orchard environment is made progressively more difficult.

Several UAV studies have used reinforcement learning for navigation, obstacle avoidance, target search, or trajectory design. Memory-based DRL has been shown to help UAV obstacle avoidance under limited environment knowledge by retaining relevant temporal information [ref]. Safety-aware DRL with moving-object prediction has been proposed to assign safety values to candidate positions and support collision-free flight [ref]. Q-learning with weighted priorities has been used to balance path length, safety, and energy in global UAV navigation [ref]. Multi-UAV adaptive path planning studies further show that DRL can optimize informative data collection and cooperative behavior [ref]. These works motivate our use of recurrent memory and safety values, but most of them do not study demand-aware spraying or the interaction between spray activation, safety control, and mission export.

SUBSECTION: Simulation-to-Mission Robotics Tools

Gazebo and ROS-based simulation tools are widely used for robotics development and evaluation [ref]. In agricultural UAV spraying, a grid-level policy is only useful if its path can be translated into a form executable by a mission manager. Therefore, this work explicitly exports learned paths as AAS mission YAML files, which is a necessary step toward validating that the learned planner is not merely an offline grid artifact.

SUBSECTION: Positioning of This Work

Compared with optimization-based pesticide irrigation [ref], crop and weed localization pipelines [ref], and UAV safety/navigation DRL studies [ref], this paper studies a combined problem: a UAV must satisfy variable spray demand, avoid static and moving obstacles, reduce redundant spraying, and produce an executable mission artifact. The key design choice is hierarchical decomposition. Instead of forcing a DQN policy to learn motion, spray timing, and safety at once, we compare both explicit and decomposed action spaces and show that automatic spraying plus safety control changes the learning behavior substantially.

SECTION: Problem Formulation

SUBSECTION: Environment Model

The environment is derived from an AAS orchard world. The parser extracts enabled tree-row models, static obstacle models, and optional field-region targets from the simulator assets. The planning layer discretizes the environment into a grid. Each target tree or field cell becomes a spraying target, while obstacle-inflated cells are marked as invalid. The default local tree-row task contains 21 target trees and 8 static obstacles with a 5 m grid resolution. Larger field and block modes reuse the same Gazebo world but change the target mask. Table~tab:env_stats summarizes the environment settings used in the current experimental record.

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TABLE/FIGURE: AAS-derived orchard planning environments.

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Setting	Targets	Obstacles	Grid	Limit
Static tree-row	21	8 static	5 m	500 steps
Variable-rate tree-row	21	8 static	5 m	500 steps
Enhanced dynamic	21	8 static + 2 dynamic	5 m	500 steps
Field rectangle	126	scene mask	10 m spacing	220 waypoints
Whole-farm blocks	block mask	masked scene	50 m spacing	1500--2000 steps

SUBSECTION: MDP Definition

We formulate spraying path planning as a Markov decision process

$$M = (S, A, P, R,),$$

where S is the state space, A is the action space, P is the transition function, R is the reward function, and is the discount factor.

The compact observation includes normalized UAV position, the nearest unsatisfied target, relative target direction, coverage progress, local obstacle indicators, demand satisfaction information, and dynamic safety features when dynamic obstacles are enabled. The exact feature set is task-dependent, because static coverage, variable-rate spraying, and dynamic-obstacle settings expose different signals.

SUBSECTION: Action Spaces

We evaluate two action designs. The 4-action space contains only grid motion:

```
A_4 = , down, left, right\.
```

In this setting, spraying can be handled by a rule controller. The 8-action space combines motion with spray activation:

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A_8 = A_4 \ on, spray\ off\.
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The 8-action setting is more expressive but harder to explore, since the agent must learn navigation and spray timing jointly.

SUBSECTION: Reward Function

The reward combines step cost, target progress, demand satisfaction, redundant spraying, collision penalty, safety penalty, and task completion:

$R_t = r_{step} + r_{new} + r_{demand} + r_{repeat} + r_{collision} + r_{safety} + r_{finish}$.

Newly covered or newly satisfied targets receive positive reward, while repeated spraying and invalid movement are penalized. In dynamic-obstacle settings, a normalized safety value $S_v[0,1]$ is computed from the UAV's distance to moving obstacles. Higher S_v indicates safer motion.

For the basic coverage experiments, the main reward coefficients are $r_{step}=-0.01$, $r_{new}=5.0N_{new}$, $r_{repeat}=-0.15N_{repeat}$, $r_{collision}=-2.0$, and $r_{finish}=50.0$. In variable-rate spraying, the coverage term is augmented with demand satisfaction so that progress is measured by satisfied dose rather than only by visiting a tree once.

SECTION: Hierarchical Spraying Architecture

SUBSECTION: Motivation

Initial experiments showed that directly learning both movement and spray switching in the 8-action enhanced task is unstable. Increasing training steps alone did not resolve the issue. We therefore introduce a hierarchical design that separates three responsibilities: RL path selection, demand-aware spray activation, and local safety override.

SUBSECTION: Automatic Spray Controller

The automatic spray controller turns spraying on when the current spray radius contains targets with unsatisfied demand, and turns it off when local demand has already been met. This converts the agent's main decision back to motion selection and avoids wasting exploration on trivial spray-on/off decisions.

SUBSECTION: Safety Controller

The safety controller checks whether the proposed action would cause a boundary violation, static obstacle collision, or predicted dynamic obstacle collision. If so, it locally overrides the action with a safe alternative when one exists. This mechanism is not intended to replace learning, but to enforce a simple safety layer around the learned policy.

SUBSECTION: DRQN and Curriculum Guidance

DRQN augments DQN with a recurrent hidden state:

$h_t = GRU(f(s_t),h_{t-1}),\ Q_t=g(h_t)$, where f encodes the current observation and g maps the hidden state to Q-values. This is useful when local observations do not fully capture previous coverage, demand residuals, or dynamic obstacle trends.

For whole-farm multi-block coverage, we also evaluate curriculum-guided exploration. During early training, the agent follows a shortest-path expert action with probability

p_t = p_0(1-_guide),

and then gradually transitions to its own value policy. The expert is used only during training, not as an evaluation-time controller.

SECTION: Experimental Setup

SUBSECTION: Algorithms

We evaluate DQN, Double DQN, Dueling DQN, Rainbow DQN lite, PPO, and DRQN in the staged experiments. DQN is the simplest value baseline. Double DQN is used to test whether separating action selection and target-value estimation helps static coverage. Dueling DQN is a strong static-coverage baseline. Rainbow DQN lite combines selected DQN improvements, but does not include the full Rainbow set such as distributional value learning and NoisyNet exploration. PPO is included as a policy-gradient reference in the static screening stage. DRQN is used to test whether recurrent memory improves dynamic and partially observable spraying tasks.

SUBSECTION: Baselines

Table~tab:baseline_roles summarizes the baseline set. The traditional baselines are not treated as weak strawmen: they are used to test whether the orchard task can already be solved by target ordering or row-wise traversal. In the whole-farm block setting, the nearest-target and row-wise baselines both cover all target cells, but the path lengths differ by 3100 m, which shows that ordering and traversal structure are important even before learning is introduced.

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TABLE/FIGURE: Baselines and their role in the study.

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Baseline & Purpose & Current evidence \\
Random policy & Sanity lower bound & used for debugging, not a headline result \\
Greedy nearest-target & Local target ordering & whole-farm 100.0%, 4640.0 m \\
Row-wise/lawnmower & Classical coverage path & field 1333.6 m; blocks 7740.0 m \\
Rule auto-spray + heuristic path & Non-learning spraying check & isolates spray logic from motion learning \\
DQN-family & Learned motion policies & static, enhanced, and whole-farm trials \\
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SUBSECTION: Tasks and Metrics

The experiments are organized into four levels:

Static tree-row coverage.

Variable-rate spraying with intelligent irrigation demand.

Enhanced dynamic-obstacle spraying with auto-spray and safety control.

Whole-farm multi-block coverage with sparse targets and curriculum guidance.

This staged design follows the diagnostic structure of the full study. The static task verifies whether value-based policies can learn basic coverage and static obstacle avoidance. The variable-rate task tests whether dose demand can be represented in the reward and observation. The dynamic enhanced task adds moving obstacles and S_v safety values. The whole-farm task expands the target mask from a single tree row to multiple sparse agricultural blocks.

We report success rate, coverage, demand satisfaction, path length, repeated spraying, planning collisions, dynamic collisions, minimum S_v, and waypoint count. In intelligent-irrigation tasks, success is defined by demand satisfaction. In the current enhanced-task protocol, the threshold is 90% demand satisfaction with zero planning collisions. Unless otherwise stated, enhanced-task episodes have a maximum of 500 steps.

SUBSECTION: Statistical Protocol

The main hierarchical enhanced experiment uses seeds 7, 11, 19, 23, and 31, 15k training steps, 2 corridor-mode dynamic obstacles, auto-spray, safety control, and intelligent irrigation. Ablations use the same five seeds and 10k training steps. Whole-farm guided DRQN uses three seeds and 20k training steps. Results are reported as mean +/- sample standard deviation across completed seeds.

SUBSECTION: Implementation Environment

Experiments were run on a Windows workstation with WSL2. The CPU is a 13th Gen Intel Core i7-13650HX with 20 logical processors, and the GPU is an NVIDIA GeForce RTX 4060 Laptop GPU with 8 GB memory and driver version 596.21. The Python environment uses Python 3.12.3, PyTorch 2.11.0, Gymnasium 1.2.3, Stable-Baselines3 2.8.0, NumPy 2.4.4, and Matplotlib 3.10.8. The implementation is based on the AAS repository revision 2dcfc26. For double-blind submission, local usernames, absolute paths, and repository-specific personal identifiers are omitted.

SECTION: Results

SUBSECTION: Static Tree-Row Coverage

The first experiment evaluates whether DQN-family algorithms can solve the local tree-row spraying task before adding demand and dynamic obstacles. Table~tab:static_tree reports three-seed results for five RL algorithms. Dueling DQN, Double DQN, DQN, and Rainbow DQN lite all reach 66.7% success, while PPO reaches 33.3%. Dueling DQN and Double DQN obtain the highest average coverage, indicating that DQN variants are suitable static baselines before the enhanced task is introduced.

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TABLE/FIGURE: Static tree-row coverage results over three seeds.

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Algorithm & Success & Coverage & Path (m) & Coll. \\
Dueling DQN & 66.7 & 92.1 +/- 13.7 & 1093.3 +/- 1218.6 & 0.0 +/- 0.0 \\
Double DQN & 66.7 & 92.1 +/- 13.7 & 1110.0 +/- 1203.8 & 0.0 +/- 0.0 \\
DQN & 66.7 & 84.1 +/- 27.5 & 1106.7 +/- 1206.7 & 0.0 +/- 0.0 \\
Rainbow lite & 66.7 & 81.0 +/- 33.0 & 1090.0 +/- 1221.4 & 0.0 +/- 0.0 \\
PPO & 33.3 & 54.0 +/- 39.9 & 1793.3 +/- 1224.0 & 0.0 +/- 0.0 \\
```

Successful episodes have much shorter paths than the aggregate averages because failed seeds run until the step limit. In successful episodes, Dueling DQN has 390.0 +/- 42.4 m path length, Double DQN has 415.0 +/- 7.1 m, DQN has 410.0 +/- 0.0 m, and Rainbow DQN lite has 385.0 +/- 35.4 m. This motivated using Dueling DQN as the static strong baseline and using DQN as the simplest interpretable value baseline.

SUBSECTION: Enhanced Task Before Hierarchical Control

The next experiments add variable-rate demand, dynamic obstacles, and explicit spray switching. Under a 97% demand threshold with 30k training steps, Rainbow DQN lite and DRQN each succeed in only one of five seeds, while DQN and Dueling DQN do not reach the threshold. A stricter 100k-step 8-action corridor-obstacle configuration also fails to stabilize: DRQN obtains 41.7% +/- 35.4% demand satisfaction and Rainbow DQN lite obtains only 10.1% +/- 14.9%, with all episodes reaching the maximum path length. These failed-but-diagnostic results show that simply increasing the action space and training budget is not enough; spray switching makes exploration substantially harder. This diagnosis motivates the hierarchical auto-spray and safety design evaluated next.

SUBSECTION: Hierarchical Enhanced Task

Table~tab:main5seed shows the five-seed enhanced-task results. DQN achieves the highest success rate, while DRQN obtains the highest average demand satisfaction and lower variance. All four algorithms report zero planning and dynamic collisions.

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TABLE/FIGURE: Five-seed hierarchical enhanced task results.

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Algorithm	Success	Coverage	Demand	Path (m)	Plan Coll.	Dyn. Coll.
DQN	80.0	82.9 +/- 30.4	78.7 +/- 29.3	844.0 +/- 926.5	0.0 +/- 0.0	0.0 +/- 0.0
DRQN	60.0	83.8 +/- 15.6	80.9 +/- 14.7	1270.0 +/- 1123.2	0.0 +/- 0.0	0.0 +/- 0.0
Dueling DQN	60.0	57.1 +/- 52.2	55.0 +/- 50.2	1244.0 +/- 1147.1	0.0 +/- 0.0	0.0 +/- 0.0
Rainbow lite	20.0	63.8 +/- 23.7	60.2 +/- 24.4	2092.0 +/- 912.3	0.0 +/- 0.0	0.0 +/- 0.0

These aggregate path lengths include failed episodes that reach the step limit. Among successful trials only, the corresponding path lengths are 430.0 +/- 43.2 m for DQN, 450.0 +/- 40.0 m for DRQN, 406.7 +/- 50.3 m for Dueling DQN, and 460.0 m for Rainbow DQN lite. These results suggest that the hierarchical design makes the enhanced task learnable even with short training budgets. DQN is a strong baseline when spraying and safety are handled by rule layers, whereas DRQN remains attractive because it has the highest mean demand satisfaction and is better aligned with partial observability.

SUBSECTION: Ablation Study

Table~tab:ablation reports the best-performing algorithm in each ablation setting. Auto-spray is the largest contributor to performance. With 4 actions and no safety, switching from always-spray to auto-spray increases the best success rate from 0.0% to 40.0%. With safety enabled, auto-spray reaches 60.0% success. The safety controller primarily removes collisions: the auto-spray no-safety setting has nonzero collision counts in the detailed logs, while the corresponding safety setting reports zero collisions.

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TABLE/FIGURE: Ablation results over five seeds. Each row reports the best algorithm for that configuration.

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Actions	Spray	Safety	Best	Success	Coverage	Demand	Coll.
4	always	no	DRQN	0.0	52.4 +/- 33.8	48.2 +/- 31.5	0.0 +/- 0.0
4	auto	no	DRQN	40.0	88.6 +/- 14.9	86.0 +/- 13.7	0.6 +/- 0.5

```
4 & always & yes & DQN & 40.0 & 64.8 +/- 42.4 & 59.8 +/- 40.7 & 0.0 +/- 0.0 \\
4 & auto & yes & DQN & 60.0 & 75.2 +/- 29.8 & 72.0 +/- 29.4 & 0.0 +/- 0.0 \\
8 & explicit & no & DRQN & 20.0 & 91.4 +/- 11.4 & 88.1 +/- 9.1 & 0.8 +/- 0.4 \\
8 & explicit & yes & DRQN & 60.0 & 75.2 +/- 27.8 & 72.1 +/- 27.6 & 0.0 +/- 0.0 \\
```

The 8-action setting is more expressive, and DRQN can achieve high demand satisfaction in it, but it is less stable and more collision-prone without safety. This supports the design choice of keeping the RL action space small while delegating spraying to an interpretable demand-aware controller.

SUBSECTION: Learning Curves

We log periodic deterministic evaluations during training. Fig.~fig:learning_curves shows reward, coverage, and demand satisfaction curves for the auto-spray+safety ablation, where all five seeds use the same curve-logging protocol. The curves indicate that the hierarchical controller improves the training signal: coverage and demand satisfaction rise early, while variance across seeds remains visible. The five-seed enhanced-task table remains the primary result for the main configuration; main-configuration learning dynamics are excluded from the figure because not all main runs were logged under the same curve protocol.

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[Figure file: outputs/paper_required_experiments/ablation_autospray_safety/summary/learning_curves.png]

TABLE/FIGURE: Training curves for the auto-spray+safety ablation: reward, coverage, and demand satisfaction.

SUBSECTION: Whole-Farm Multi-Block Coverage

The AAS orchard world is not limited to the enabled tree row. We also define a rectangular field mode and a multi-block mode using target masks in the same world. In field mode, field-bounds=(-55,-105,75,-20) and 10 m spacing generate 126 abstract field targets; a lawnmower mission covers 100.0% of these field targets with about 1333.6 m path length and zero collisions. In block mode, roads, parking regions, buildings, and non-farm textures are excluded from the spraying target mask, but the UAV may still fly over non-target areas when transferring between separated blocks.

Before applying RL to the whole-farm setting, we verify the target mask with two rule baselines. A nearest-target baseline covers 100.0% of target cells with a 4640.0 m path, while a row-wise lawnmower baseline also covers 100.0% but requires 7740.0 m. This confirms that block order and traversal pattern strongly affect path efficiency.

When DQN-family methods are directly applied to the larger blocks target mode without guidance, exploration fails. In a short smoke test, the best coverage is only 15.9% by Rainbow DQN lite, followed by 13.4% by Dueling DQN; DRQN and DQN remain near local regions. After adding curriculum-guided exploration, DRQN reaches 100.0% success and 97.56% +/- 0.00% coverage over three seeds, with 4425.0 +/- 81.9 m average path length, zero collisions, zero repeated spraying, and 886.0 +/- 16.4 waypoints.

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TABLE/FIGURE: Whole-farm block coverage results.

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```
Method & Success & Coverage & Path (m) & Coll. \\
Greedy nearest-target & 100.0 & 100.0 & 4640.0 & 0.0 \\
Row-wise lawnmower & 100.0 & 100.0 & 7740.0 & 0.0 \\
```

```
Unguided Rainbow lite & 0.0 & 15.9 & 10000.0 & 0.0 \\
Curriculum-guided DRQN & 100.0 & 97.56 +/- 0.00 & 4425.0 +/- 81.9 & 0.0 +/- 0.0 \\
```

This result does not imply that pure end-to-end RL solves whole-farm spraying. Rather, it supports a hierarchical interpretation: a global prior or curriculum is needed to handle sparse long-distance exploration, while DRQN provides the local sequential decision module.

We visualize learned and rule-based trajectories using square action-matrix plots. These plots are generated directly from path summary files and show targets, covered cells, obstacles, start position, and discrete motion directions. The visualization helps separate simulator-scale limitations from task-definition limitations: in the local tree-row task, policies remain local because only the enabled tree row is marked as the target, not because the Gazebo world is small.

SECTION: Mission Export and Gazebo Replay Validation

To connect the offline grid planner to the robotics simulator, we export successful RL paths into AAS mission YAML files. A generated mission contains takeoff, wait, speed-setting, waypoint reposition, and landing commands that can be executed by the AAS mission node. The current system has produced missions for static DQN/Rainbow examples, hierarchical DRQN tree-row spraying, rectangular field lawnmower coverage, whole-farm block coverage, and the paper replay case.

SUBSECTION: Mission Export Case

The paper replay case is generated from a DRQN run in the hierarchical enhanced task. Table~tab:gazebo_replay_case reports the planning-layer metrics for the exported path. The generated mission is spray_paper_replay.yaml.

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TABLE/FIGURE: Planning-layer metrics for the exported replay mission.

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Metric & Value \\
Algorithm / seed & DRQN / 7 \\
Coverage & 95.2% \\
Demand satisfaction & 90.9% \\
Path length & 410.0 m \\
Planning collisions & 0 \\
Dynamic collisions & 0 \\
Minimum S_v & 0.33 \\
Mission waypoints & 83 \\
Mission file & spray_paper_replay.yaml \\
```

SUBSECTION: Replay Evidence Record

The mission export result verifies path-to-mission conversion. A complete simulator validation record must also show that the exported mission executes in AAS/Gazebo without collision. Table~tab:replay_evidence lists the required replay artifacts. In the present experimental record, these live Gazebo/QGroundControl artifacts have not yet been collected, so the paper should not claim completed simulator replay until these rows are replaced by actual screenshots, logs, and trajectory plots.

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TABLE/FIGURE: Gazebo/QGroundControl replay validation record.

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Evidence item & Expected artifact & Current status \\

Gazebo world launch & apple-orchard screenshot & pending \\
QGroundControl loaded/connected & QGC screenshot & pending \\
Takeoff and waypoint execution & mission node log & pending \\
Landing or mission completion & mission/QGC log & pending \\
ROS 2 position trajectory & position-topic CSV or rosbag & pending \\
No-collision execution & screenshot or trajectory audit & pending \\
Grid path vs replay trajectory & comparison figure & pending \\
```

SECTION: Discussion

The experiments support four observations. First, DQN remains a strong baseline when the task is hierarchically simplified; therefore it should not be dismissed. Second, Dueling DQN is a stable static tree-row baseline, but its advantage does not automatically transfer to the dynamic demand-aware setting. Third, DRQN is better suited to settings that contain partial observability, explicit spray switching, or sparse whole-farm exploration. Fourth, auto-spray and safety control are not minor engineering details: they change the learning problem by separating spraying and safety from motion selection.

The main limitation is that dynamic obstacles and variable-rate demand are modeled in the high-level Python planning environment rather than as full physical entities in Gazebo. The current spray model also does not simulate droplet physics, wind drift, deposition, or tank constraints. In addition, the enhanced-task success threshold is currently 90% demand satisfaction; stronger claims about high-precision spraying require longer training or curriculum experiments at a 97% demand threshold. The whole-farm guided result depends on early expert actions, and should be interpreted as evidence for a global-prior-plus-local-RL architecture rather than as pure end-to-end reinforcement learning. Finally, the current artifact proves mission export, while live Gazebo/QGroundControl replay evidence remains required before making a completed simulator-execution claim.

SECTION: Conclusion

We presented a hierarchical DQN-family framework for safe variable-rate UAV spraying in an AAS orchard simulator. The central result is that the task is more effectively handled as a decomposition of motion learning, demand-aware automatic spraying, and local safety control than as a single end-to-end DQN problem. The experiments show a clear progression: Dueling DQN is a stable static baseline; explicit 8-action spraying is difficult to train; auto-spray and safety control substantially improve enhanced-task learnability and safety; and curriculum-guided DRQN is needed for sparse whole-farm coverage. The framework also exports learned paths to AAS mission YAML, providing the interface required for full Gazebo replay validation.